DSAI510 Assignment 09

December 9, 2025

0.1 Assignment 9 - Deadline: Dec 14, 2025, Sun 11pm

DSAI 510 Fall 2025 Complete the assignment below and upload both the .ipynb file and its pdf to https://moodle.boun.edu.tr by the deadline given above. The submission page on Moodle will close automatically after this date and time.

To make a pdf, this may work: Hit CMD+P or CTRL+P, and save it as PDF. You may also use other options from the File menu.

[1]:
```
# Run this cell first

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import statsmodels.api as sm

# Set the display option to show all rows scrolling with a slider
# pd.set_option('display.max_rows', None)
# To disable this, run the line below:
# pd.reset_option('display.max_rows')
```

0.2 Note:

In the problems below, if they ask "show the number of records that are nonzero", the answer is a number; so you don't need to show the records themselves. But if it asks, "show the records with NaN", it wants you to print those records (rows) containing NAN and other entries, not asking how many such records there are. So be careful about what you're asked.

1 Problem 1: Newsgroup discussions classification (30 pts)

The `fetch_20newsgroups` dataset is a collection of approximately 20,000 newsgroup documents, distributed across 20 different newsgroups. Each newsgroup represents a distinct category, ranging from technology and politics to religion and sports. This dataset is widely used in natural language processing and machine learning for tasks like text classification and clustering, due to its diverse range of topics and real-world discussion content.

Your goal is to build a classifier that would take a document from this dataset and output the category of that document. Categories: 'sci.med', 'comp.graphics', 'sci.electronics' and 'talk.politics.misc'.

The code to load the data is already provided below, and the first five documents and their class labels are printed.

1) Use TF-IDF to vectorize the train and test datasets.

(Important for preventing data leakage: Remember from the the Jupyter notebook we used in the lecture; you use `fit_transform()` on the training dataset to learn the vocabulary (the list of words), and on the test dataset, you use only `transform()` to represent vectors of the test documents in terms of vocabulary learned from the training dataset. It is likely that there will be words that appear only in the test dataset and not in the training dataset. Such words will simply be ignored when the model is applied to the test dataset for evaluation, because the vocabulary is learned from the training dataset via `fit_transform()`.)

2) Use a classifier of your choice to train your model using the TD-IDF vectors of documents. Then, calculate the accuracy, recall, F1 and precision of your model by using the test data.

(Note that we have four classes here, so your classifier will not be a binary classifier but a multi-class classifier. This means your model should be capable of distinguishing and predicting among four different categories.)

3) Find one document from the test data for each categories: 'sci.med', 'comp.graphics', 'sci.electronics' and 'talk.politics.misc'. Print those four documents and print their category as a string (not as a number, but like 'sci.med' etc.).

[2]:
```
from sklearn.datasets import fetch_20newsgroups
# Selected categories
categories = ['sci.med', 'comp.graphics', 'sci.electronics', 'talk.politics.
 ↪misc']

# Loading training dataset
df_train = fetch_20newsgroups(subset='train',categories=categories,␣
 ↪shuffle=True, random_state=3)

# Loading testing dataset
df_test = fetch_20newsgroups(subset='test',categories=categories)
```

[3]:
```
type(df_train)
```

[3]:
```
sklearn.utils._bunch.Bunch
```

[4]:
```
df_train.keys()
```

[4]:
```
dict_keys(['data', 'filenames', 'target_names', 'target', 'DESCR'])
```

[11]:
```
df_train.target
```

[11]:
```
array([1, 2, 1, …, 1, 0, 3], shape=(2234,))
```

[8]:
```
# First four documents as examples
```

```
for i in range(5):
    print(f"\033[1m\033[91mDocument {i}:\033[0m ")  # Red and bold text
    print("Target:", df_train.target[i])
    print(df_train.data[i])
    print("\n------------------\n")  # Separator for readability
```

```
Document 0:
Target: 1
From: peter.m@insane.apana.org.au (Peter Tryndoch)
Subject: Dmm Advice Needed
Lines: 28

AllMartin EmdeDMM Advice Needed

ME>From: mce5921@bcstec.ca.boeing.com (Martin Emde)
ME>Organization: Boeing
ME>
ME>I an currely in the market for a DMM and recently saw an add
ME>for a Kelvin 94 ($199).  Does anyone own one of these or some
ME>other brand that they are extremely happy with.  How do the
ME>small name brands compare with the Fluke and Beckman brands?
ME>I am willing to spend ~$200 for one.
ME>
ME>Any help is greatly appreciated. (please email)
ME>
ME>-Martin

If you are going to use one where it counts (eg:aviation, space scuttle,
etc) then I suggest you go and buy a Fluke (never seen a Beckman), however
for every other use you can buy a cheapie. I have a metex which is some
made up name, as I have seen the same DMM with other brand names on it, I
bought it about 4 yrs ago for Aus$125.00 (convert that to US and you see
that it's definetly a cheapie.) So far it has proved to be accurate, taken
moderate abuse, and has many features on it (CAP, FREQ,Transistor check,
etc). I am very happy with it and would definetly not buy a fluke just for
the name. Hope this helps.

Cheers
Peter T.



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Document 1:
Target: 2
From: bmdelane@quads.uchicago.edu (brian manning delaney)
Subject: Re: diet for Crohn's (IBD)
```

Reply-To: bmdelane@midway.uchicago.edu
Organization: University of Chicago
Lines: 27

One thing that I haven't seen in this thread is a discussion of the
relation between IBD inflammation and the profile of ingested fatty
acids (FAs).

I was diagnosed last May w/Crohn's of the terminal ileum. When I got
out of the hospital I read up on it a bit, and came across several
studies investigating the role of EPA (an essentially FA) in reducing
inflammation. The evidence was mixed. [Many of these studies are
discussed in "Inflammatory Bowel Disease," MacDermott, Stenson. 1992.]

But if I recall correctly, there were some methodological bones to be
picked with the studies (both the ones w/pos. and w/neg. results). In
the studies patients were given EPA (a few grams/day for most of the
studies), but, if I recall correctly, there was no restriction of the
other FAs that the patients could consume. From the informed
layperson's perspective, this seems mistaken. If lots of n-6 FAs are
consumed along with the EPA, then the ratio of "bad" prostanoid
products to "good" prostanoid products could still be fairly "bad."
Isn't this ratio the issue?

What's the view of the gastro. community on EPA these days? EPA
supplements, along with a fairly severe restriction of other FAs
appear to have helped me significantly (though it could just be the
low absolute amount of fat I eat -- 8-10% calories).

-Brian <bmdelane@midway.uchicago.edu>

Document 2:
Target: 1
From: johnh@macadam.mpce.mq.edu.au (John Haddy)
Subject: Re: Help with ultra-long timing
Organization: Macquarie University
Lines: 39
Distribution: world
NNTP-Posting-Host: macadam.mpce.mq.edu.au

In article <C513wI.G5A@athena.cs.uga.edu>, mcovingt@aisun3.ai.uga.edu (Michael
Covington) writes:
|> (1) Don't use big capacitors. They are unreliable for timing due to
|> leakage.

|>
|> Instead, use a quartz crystal and divide its frequency by 2 40 times
|> or something like that.
|>
|> 1 MHz divided by 2^40 = 1 cycle per 2 weeks, approximately.
|>
|> (2) I wouldn't expect any components (other than batteries or electrolytic
|> capacitors) to fail at -40 C (or -40 F for that matter either :)).
|> The battery is going to be your big problem. Also, of course, your
|> circuit shouldn't depend on exact values of resistors (which a crystal-
|> controlled timer won't).
|>

… Wouldn't a crystal be affected by cold? My gut feeling is that, as a
mechanically resonating device, extreme cold is likely to affect the
compliance (?terminology?) of the quartz, and hence its resonant frequency.

```
|> --
|> :-  Michael A. Covington         internet mcovingt@ai.uga.edu :    *****
|> :-  Artificial Intelligence Programs       phone 706 542-0358 :  *********
|> :-  The University of Georgia                fax 706 542-0349 :   *  *  *
|> :-  Athens, Georgia 30602-7415 U.S.A.     amateur radio N4TMI :  ** *** **
```

JohnH

```
----------------------------------------------------------------------------

      |  _  |_   _   |_|  _   _|  _|             Electronics Department
    |_| (_) | | | |  | | (_| (_| (_| \/          School of MPCE
    ---------------------------------/-          Macquarie University
                                                 Sydney, AUSTRALIA 2109

    Email: johnh@mpce.mq.edu.au, Ph: +61 2 805 8959, Fax: +61 2 805 8983

----------------------------------------------------------------------------
```

Document 3:
Target: 3
From: bob1@cos.com (Bob Blackshaw)
Subject: Re: Tieing Abortion to Health Reform -- Is Clinton Nuts?
Organization: Corporation for Open Systems
Distribution: world
Lines: 37

In <1993Apr5.170349.10700@ringer.cs.utsa.edu> sbooth@lonestar.utsa.edu (Simon E. Booth) writes:

>In article <1993Apr2.230831.18332@wdl.loral.com> bard@cutter.ssd.loral.com
writes:
>>sbooth@lonestar.utsa.edu (Simon E. Booth) writes:
>># sandvik@newton.apple.com (Kent Sandvik) writes:
>># >We already kill people (death penalty), and that costs even more
>># >money, so you could as well complain about this extremely barbaric
>># >way of justice.
>>#
>># But the death penalty is right.
>>#
>># And how expensive can an execution be? I mean, I think rope, cyanide
>># (for the gas), or the rifles and ammunition to arm firing squads are
>># affordable.
>>#
>># Now, perhaps lethal injection might be expensive, in that case, let's
>># return to the more efficient methods employed in the past.
>>
>>Oh, sure, the death *penalty* is fairly inexpensive, but the trial and
>>sentencing can run millions.

>>
>>--strychnine unless you wanna cut costs by skipping the trial and
>> sentencing… you murderous little rat-bastard

> Why as a matter of fact, I was thinking of that as a way to make
>the system more efficient. And the only murderous rat-bastards are
>aboritionists.

Yeah, Simon's no rat-bastard, he's the Head Attack Puppy :-)

>Simon

TOG

Document 4:
Target: 3
From: rscharfy@magnus.acs.ohio-state.edu (Ryan C Scharfy)
Subject: Re: Good Neighbor Political Hypocrisy Test
Nntp-Posting-Host: magnusug.magnus.acs.ohio-state.edu

Organization: The Ohio State University
Lines: 59

In article <1993Apr16.141409.25036@pmafire.inel.gov> cdm@pmafire.inel.gov (Dale
 Cook) writes:
>In article <1993Apr15.193603.14228@magnus.acs.ohio-state.edu> rscharfy@magnus.
acs.ohio-state.edu (Ryan C Scharfy) writes:
>>In article <stevethC5JGCr.1Ht@netcom.com> steveth@netcom.com (Steve Thomas) w
ri
>>tes:
>>
>>>Just _TRY_ to justify the War On Drugs, I _DARE_ you!
>>
>>A friend of mine who smoke pot every day and last Tuesday took 5 hits of acid

>>is still having trouble "aiming" for the bowl when he takes a dump. Don't as

>>me how, I just have seen the results.
>>
>>Boy, I really wish we we cut the drug war and have more people screwed up in
>>the head.
>
>I'm sorry about your friend. Really. But this anecdote does nothing to
>justify the "war on drugs". If anything, it demonstrates that the "war"
>is a miserable failure. What it demonstrates is that people will take
>drugs if they want to, legal or not. Perhaps if your friend were taking
>legal, regulated drugs under a doctors supervision he might not be in the
>position he's in now.
>

I do agree with you, in a way. The war on drugs has failed, but in my opinion,
that doesn't mean we have to give up. Only change the tactics.

For instance, here are how some penalties should be changed.

Dealing Coke -- Death
Dealing Heroin -- Death
Dealing Pot -- Death
Dealing Crack -- Death

The list goes on and on!!!…

JUST KIDDING!!!

However, on a more serious note, I do believe that we should take some money
away from the foriegn operations in South America and costly border
interdiction efforts. (Don't think I'm going to say, "spend it to educate
people", because I know plenty of educated dopers). Actually, spend it on

```
things like drug treatment programs.

I saw an interesting story on 60 minutes about how the British actually
prescribe and addict his "recommended" dosage, and try to ween him off from it,
or cut the amount down to levels where it is "acceptable".  Sounds good so far
from what I heard with a decrease in cost, lower addiction rates by wiping out
the dealer's markets, etc. (But that was the only thing I have heard about it.)

However, legalizing it and just sticking some drugs in gas stations to be
bought like cigarettes is just plain silly.  Plus, I have never heard of a
recommended dosage for drugs like crack, ecstasy, chrystal meth and LSD.
The 60 Minute Report said it worked with "cocaine" cigarettes, pot and heroin.

Ryan


------------------
```

1.1 1

[ ]:
```python
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer()
X_train = vectorizer.fit_transform(df_train.data)
X_test = vectorizer.transform(df_test.data)

print(f"Training data: {X_train.shape}")
print(f"Test data: {X_test.shape}")
```

```
Training data: (2234, 36433)
Test data: (1488, 36433)
```

1.2 2

[15]:
```python
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

clf = LogisticRegression(random_state=42, max_iter=1000)
clf.fit(X_train, df_train.target)
y_pred = clf.predict(X_test)

acc = accuracy_score(df_test.target, y_pred)
print(f"Overall Accuracy: {acc:.4f}\n")

print("Classification Report:")
print(classification_report(df_test.target, y_pred, target_names=categories))
```

```
Overall Accuracy: 0.8898
```

```
Classification Report:
                    precision    recall  f1-score   support

           sci.med       0.85      0.92      0.88       389
     comp.graphics       0.86      0.91      0.88       393
   sci.electronics       0.93      0.85      0.89       396
talk.politics.misc       0.94      0.89      0.92       310

          accuracy                           0.89      1488
         macro avg       0.89      0.89      0.89      1488
      weighted avg       0.89      0.89      0.89      1488
```

1.3 3

[17]:
```python
import numpy as np

target_categories = ['sci.med', 'comp.graphics', 'sci.electronics', 'talk.
 ↪politics.misc']

for category_name in target_categories:

    category_id = df_test.target_names.index(category_name)
    matching_indices = np.where(df_test.target == category_id)[0]

    sample_index = matching_indices[0]

    document_text = df_test.data[sample_index]

    print(f"Category String: {category_name}")
    print(f"Target Label ID: {category_id}")
    print(f"Test Set Index : {sample_index}")
    print("-"*40)
    print(document_text)
    print("="*80 + "\n")
```

```
Category String: sci.med
Target Label ID: 2
Test Set Index : 11
----------------------------------------
Organization: Arizona State University
From: <ICBAL@ASUACAD.BITNET>
Subject: Re: Opinions on Allergy (Hay Fever) shots?
Distribution: world
 <93115.120409ICBAL@ASUACAD.BITNET> <1rhb0e$9ks@europa.eng.gtefsd.com>
Lines: 13
```

In article <1rhb0e$9ks@europa.eng.gtefsd.com>, draper@gnd1.wtp.gtefsd.com (PAM
DRAPER) says:
>
>This homeopathic remedies. I tried the dander one for a month. 15 drops
>three times a day. I didn't notice any change whats so ever. How long
>were you using the drops before you noticed a difference?
>
It is NOT a homeopathic remedy. Improvement began in a few months.
I am allergic to bermuda grass and if anyone nearby was mowing a lawn
my nose would start to run. Now I can walk right by and it doesn't bother
me at all. The same success with desert ragweed.

Bruce Long

==

Category String: comp.graphics
Target Label ID: 0
Test Set Index : 0

From: lee@hobbes.cs.umass.edu (Peter Lee)
Subject: Re: QuickTime performance (was Re: Rumours about 3DO ???)
<1993Apr16.212441.34125@rchland.ibm.com>
<1993Apr26.170915.15833@waikato.ac.nz>
Reply-To: lee@cs.umass.edu
Organization: Software Development Lab, UMass, Amherst
Lines: 108
In-reply-to: ldo@waikato.ac.nz's message of 26 Apr 93 05:09:15 GMT

In article <1993Apr26.170915.15833@waikato.ac.nz> ldo@waikato.ac.nz (Lawrence
D'Oliveiro, Waikato University) writes:

Path: dime!ymir.cs.umass.edu!nic.umass.edu!noc.near.net!howland.reston.ans.ne
t!usc!elroy.jpl.nasa.gov!decwrl!waikato.ac.nz!ldo
From: ldo@waikato.ac.nz (Lawrence D'Oliveiro, Waikato University)
Newsgroups: comp.multimedia,comp.graphics
Date: 26 Apr 93 05:09:15 GMT
References: <1993Mar31.074502.3590@aragorn.unibe.ch>
<1993Apr16.212441.34125@rchland.ibm.com>
Organization: University of Waikato, Hamilton, New Zealand
Lines: 67
Xref: dime comp.multimedia:6358 comp.graphics:32606

OK, with all the discussion about observed playback speeds with QuickTime,
the effects of scaling and so on, I thought I'd do some more tests.

First of all, I felt that my original speed test was perhaps less than
realistic. The movie I had been using only had 18 frames in it (it was a

version of the very first movie I created with the Compact Video compressor). I decided something a little longer would give closer to real-world results (for better or for worse).

I pulled out a copy of "2001: A Space Odyssey" that I had recorded off TV a while back. About fifteen minutes into the movie, there's a sequence where the Earth shuttle is approaching the space station. Specifically, I digitized a portion of about 30 seconds' duration, zooming in on the rotating space station. I figured this would give a reasonable amount of movement between frames. To increase the differences between frames, I digitized it at only 5 frames per second, to give a total of 171 frames.

I captured the raw footage at a resolution of 384*288 pixels with the Spigot card in my Centris 650 (quarter-size resolution from a PAL source). I then imported it into Premiere and put it through the Compact Video compressor, keeping the 5 fps frame rate. I created two versions of the movie: one scaled to 320*240 resolution, the other at 160*120 resolution. I used the default "2.00" quality setting in Premiere 2.0.1, and specified a key frame every ten frames.

I then ran the 320*240 movie through the same "Raw Speed Test" program I used for the results I'd been reporting earlier.

Result: a playback rate of over 45 frames per second.

That's right, I was getting a much higher result than with that first short test movie. Just for fun, I copied the 320*240 movie to my external hard disk (a Quantum LP105S), and ran it from there. This time the playback rate was only about 35 frames per second. Obviously the 230MB internal hard disk (also a Quantum) is a significant contributor to the speed of playback.

I modified my speed test program to allow the specification of optional scaling factors, and tried playing back the 160*120 movie scaled to 320*240 size. This time the playback speed was over 60 fps. Clearly, the poster who observed poor performance on scaled playback was seeing QuickTime 1.0 in action, not 1.5. I'd try my tests with QuickTime 1.0, but I don't think it's entirely compatible with my Centris and System 7.1…

Unscaled, the playback rate for the 160*120 movie was over 100 fps.

The other thing I tried was saving versions of the 320*240 movie with "preferred" playback rates greater than 1.0, and seeing how well they played from within MoviePlayer (ie with QuickTime's normal synchronized playback). A preferred rate of 9.0 (=> 45 fps) didn't work too well: the playback was very jerky. Compare this with the raw speed test, which achieved 45 fps with ease. I can't believe that QuickTime's synchronization code would add this much overhead: I think the slowdown was coming from the Mac system's task switching.

A preferred rate of 7.0 (=> 35 fps) seemed to work fine: I couldn't see any evidence of stutter. At 8.0 (=> 40 fps) I *think* I could see slight stutter, but with four key frames every second, it was hard to tell.

I guess I could try recreating the movies with a longer interval between the key frames, to make the stutter more noticeable. Of course, this will also improve the compression slightly, which should speed up the playback performance even more…


```
Lawrence D'Oliveiro                        fone: +64-7-856-2889
Computer Services Dept                      fax: +64-7-838-4066
University of Waikato            electric mail: ldo@waikato.ac.nz
Hamilton, New Zealand    37^ 47' 26" S, 175^ 19' 7" E, GMT+12:00
```


I'm afraid I missed the start of this thread, but there are three factors that can significantly affect QuickTime's playback speed that you may want to take into account:

(1) playback bit depth (things are fastest when you play a movie back at the bit depth it was compressed for, this is usually 8 or 16 bit, but other depths are (of course) possible).

(2) type of scaling (QT is optimized for "double size" scaling, other scaling factors hit peformance much harder).

(3) playback window position (MoviePlayer limits your window placement choices to advantagous pixel boundaries by default, I'm not sure about Premiere).

Any combination of those can radically alter playback performance. Image size is, of course, another biggie. Giving the movie player lots of RAM can also make a real difference.

Forgive me if these were mentioned earlier in the thread…

-Peter Lee


```
--
/------------------- Peter E. Lee, Software Conductor ---------------------\
|                      Specular International, Inc.                        |
|       lee@cs.umass.edu or (413) 256-1329 (H) or (413) 549-7600 (W)       |
\-------- Beauty is 24 bits deep, plus eight bits of alpha channel ----------/
```


==

Category String: sci.electronics
Target Label ID: 1
Test Set Index : 1

From: schuch@phx.mcd.mot.com (John Schuch)
Subject: Re: Self-destructing copy protection on VHS tape?
Nntp-Posting-Host: bopper2.phx.mcd.mot.com
Organization: Motorola Computer Group, Tempe, Az.
Lines: 18

In article <klp.735603389@quark> klp@doe.carleton.ca (Ka Lun Pang) writes:
>
>I borrowed a VHS tape from a friend and it has a warning in the begining saying
>that attempts to copy the tape will result in destroying the copy and the
>original. I found this unbelievable as playing and recording are two different
>processes. However, I've never seen this tape being sold anywhere so I don't
>want to take the chance even it's small.
>
>Anyone has experience in this kind of self-destructing video tapes?
>

I have always thought that if I wanted to send the Police a tape with
a ransom demand on it, or send CNN a video tape to see if they wanted
to buy it, I would place a small magnet near the take-up spool so the
tape would be erased as it was played. Who would think to check?

John

==

Category String: talk.politics.misc
Target Label ID: 3
Test Set Index : 7

From: lazlo@carina.unm.edu (Lazlo Nibble)
Subject: Re: WACO burning
Organization: Vroom Socko International Fear Club
Lines: 11
Distribution: world
NNTP-Posting-Host: carina.unm.edu

russotto@eng.umd.edu (Matthew T. Russotto) writes:

> The idea that kerosene lamps would be all over the place (with
> electricity cut off) makes sense. The idea that ramming tanks into the
> building would spill them and cause a fire makes sense.

As does the idea that a CS gas canister can get hot enough to ignite dry baled hay.

--
Lazlo (lazlo@triton.unm.edu)

==